# Christopher Atta Amponsah, GIS & Geospatial AI Analyst, GISP

Lutz, Florida, United States, , chris.attaamponsah.ta@gmail.com, attaamponsahc@usf.edu

**LINKS** www.linkedin.com/in/christopher-atta-amponsah-pgs-ggs-13408519 | github.com/MyCodeChris

**PROFILE**

GIS Professional and Geomatic Engineer with extensive years of teaching and research experience. Specialized in predictive flood modeling, leveraging machine learning and hydraulic modeling techniques. Proficient in integrating GIS for flood prediction and conducting satellite (GPS) surveys for accurate land assessments. Committed to leveraging spatial technologies to develop innovative solutions to environmental challenges.

**EDUCATION**

| | | |
|---|---|---|
| Aug 2026 — Present | PhD Geography, Environmental Science, & Policy, University of South Florida (USF) | Tampa, FL, USA |
| Aug 2025 — Present | Graduate Certificate: Artificial Intelligence, University of South Florida (USF) | Tampa, FL, USA |
| Aug 2024 — Aug 2026 | MA Geography: GIS and Spatial Analysis, University of South Florida (USF) (*3.74/4.0 GPA*) | Tampa, FL, USA |
| Aug 2024 — Aug 2026 | Graduate Certificate: Geographic Information Science, University of South Florida | Tampa, FL, USA |
| Aug 2018 — Nov 2022 | BSc Geomatic Engineering, Kwame Nkrumah University of Science and Technology<br>Graduated with First Class Honors (*3.86/4.0 GPA*) | Kumasi, Ghana |

**COURSES**

| | | |
|---|---|---|
| Aug 2026 — Present | Computer Vision; Multivariate Statistical Analysis; Seminar in Urban Envrionment | |
| Aug 2024 — Aug 2026 | Introduction to Artificial Intelligence; Multilevel Data Analysis; Programming in GIS; Advanced GIS; Water Resources GIS; Coastal Hazard & Resilience \|\| USF, Tampa. | FL, USA |
| Aug 2018 — Nov 2022 | Computer Programming; Calculus I & II; Algebra; Remote Sensing; Satellite Surveying; GIS; Photogrammetry; Hydrographic Surveying \|\| KNUST, Kumasi. | Ghana |

**EMPLOYMENT HISTORY**

Fall 2026 — Present — Tampa, FL, USA

**Instructor, GEO 3164C: Research Methods in Geography, University of South Florida**

Sole instructor for a semester-long undergraduate course on quantitative research methods and statistical computing in R.

- Design and deliver twice-weekly lectures and labs on research design, data management, and statistical analysis.
- Develop and adapt slide decks, coding assignments, and assessments within a structured Canvas environment.
- Instruct students in data frames, data visualization with ggplot2, and reproducible analysis.
- Evaluate performance through assignments, presentations, and a final exam, and provide individualized technical support.

Aug 2024 — Present — Tampa, FL, USA

**Graduate Teaching Assistant, University of South Florida (USF)**

Facilitates student learning through clear communication and assignment support. Enhances lectures by organizing activities and preparing materials to boost student engagement. Evaluates assignments and exams, delivering constructive feedback to promote academic growth and ensure grading compliance.

- Communicates effectively with students to clarify expectations.
- Organizes engaging activities for lectures to support learning.
- Provides timely feedback on GIS assignments to aid student development.

Jun 2025 — Present — St. Pete, FL, USA

**Student Assistant (Geospatial Analytics Lab), USF**

Conducting geospatial research integrating GIS, machine learning, and hydrologic modeling for flood prediction. Performing a meta-analysis of machine-learning-based flood modeling literature to identify global research trends. Developing a GIS-integrated pluvial flood model using data-driven Neural Networks (Convolutional Neural Network), Ensemble models (Random Forest, Gradient Boosting), Automated Machine Learning (AutoML), and physics-based HEC-RAS workflows.

- Integrating drainage networks and rainfall time series into modeling frameworks
- Collaborating with teams to enhance data integration processes in ArcGIS Pro and web services.
- Aiding in the update of GIS workshop materials

Nov 2022 — Mar 2024 — Kumasi, Ghana

**Teaching Assistant, Kwame Nkrumah University of Science and Technology**

Conducted technical instruction for Remote Sensing and GIS lab sessions. Served as a field instructor for GIS, GPS surveying, remote sensing, photogrammetry, land surveying, engineering surveying, and field astronomy practicals. Evaluated assignments and examinations. Guided undergraduate students in field data collection using surveying equipment. Conducted freelance cadastral surveys using survey-grade GNSS equipment and processed observations with Trimble Business Center software.

- Delivered comprehensive GIS lab instruction.
- Assisted in UAV piloting research efforts.
- Graded academic assignments effectively.
- Led field data collection initiatives.

| | | |
|---|---|---|
| Aug 2021 — Nov 2021 | **Assistant Surveyor, Natwam Surveying and Building Designs Company Ltd.**<br>Managed survey team for land surveying projects. Conducted setting-out surveys effectively. Assisted in delivering district-wide cadastral projects.<br>• Managed and coordinated field survey activities to ensure accurate data collection.<br>• Assisted in plotting cadastral and site plans using AutoCAD based on field survey data.<br>• Supported basic spatial data validation for the municipal planning office.<br>• Collected GPS measurements for boundary definition, control points, and site mapping.<br>• Integrated GPS survey data with existing plans and reference coordinates in ArcGIS software. | Offinso, Ghana |
| Jun 2019 — Sep 2019 | **Assistant Engineer, Engineering/Works Department, Offinso Municipal Assembly**<br>Conducted technical surveys and collected data for municipal procurement projects. Conducted building site inspections to ensure compliance with public space requirements. Assisted civil engineers with stadium terrain leveling to ensure accurate foundations. Collaborated with stakeholders at the Municipal Assembly to streamline road construction processes.<br>• Conducted technical GPS surveys for municipal procurement and infrastructure projects.<br>• Inspected building sites for spatial and regulatory compliance<br>• Assisted in stadium terrain leveling and elevation control using level instruments.<br>• Streamlined road construction procedures through the preparation of AutoCAD drawings. | Offinso, Ghana |

**POSITION HELD**

| | | |
|---|---|---|
| 2025 — Present | **Treasurer**: Ghanaian Students Association, University of South Florida (USF)<br>• **Financial Strategy:** Applying for the inaugural annual budget and successfully securing budget allocation for events.<br>• **Fund Management:** Maintaining precise ledgers for all university-allocated funding using strict accounting principles.<br>**Compliance:** Navigating university financial regulations to secure institutional funding and maintain active student organization status. | Tampa, FL USA |
| 2025 — Present | **Secretary**: American Association of Photogrammetry and Remote Sensing (ASPRS), University of South Florida (USF) Student Chapter | Tampa, FL USA |
| Jun 2020 — Jun 2021 | **Academic Board Member**: Association of Geomatics Engineering Students | KNUST, Ghana |

**VOLUNTEER EXPERIENCE**

| | | |
|---|---|---|
| May 2025 — Jun 2025 | Argonne National Laboratory, US Department of Energy – *Summer Internship (Online)* | USA |
| 2021 | Department coach for the college quiz competition, College of Engineering, KNUST | Kumasi, Ghana |
| 2021 | Student Academic Mentor, Department of Geomatics Engineering, KNUST | Kumasi, Ghana |

**ASSOCIATION/MEMBERSHIPS**

| | | |
|---|---|---|
| Nov 2025 | Association of American Geographers (AAG) - Student Member | USA |
| Feb 2025 | American Society of Photogrammetry and Remote Sensing (ASPRS) – Student Member | FL, USA |
| Feb 2025 | Florida Society of Geographers - Student Member | FL, USA |
| Nov 2024 | Ghana Geospatial Society (GGS) - Professional Member | Kumasi, Ghana |

**RESEARCH EXPERIENCE**

| | | |
|---|---|---|
| 2024 — 2026 | Master's Thesis:<br>• The Application of GIS-Integrated ML Models for Flood Prediction: A Global Meta- Analysis and a Comparative Study of ML, AutoML, and HEC-RAS Flood Models. | USF, FL, USA |
| 2021 — 2022 | Undergraduate Thesis:<br>Mapping Drainage Systems with Geographical Information Systems (GIS) and Drone Technology<br>Other undergraduate projects:<br>• Traffic Volume Studies - Estimating Average Daily Traffic and Annual Daily Traffic<br>• Mapping underground utilities and cavities with Ground Penetrating Radar.<br>• Geospatial assessment of waste management in the Kumasi Business District, KANKO Hub, Ghana. | KNUST, Kumasi, Ghana |

| | |
|---|---|
| **PROGRAMMING** | Python, JavaScript, R |
| **AI/ML** | *Frameworks*: Scikit-learn, TensorFlow, Keras<br>*Algorithms*: CNN, Random Forest, XGBoost, Gradient Boosting, SVM, k-NN, AutoML<br>*Data & scientific computing*: NumPy, Pandas, Matplotlib, SciPy, SAS |

| | |
|---|---|
| **TOOLS** | *Geospatial platforms*: ArcGIS Pro, QGIS, Google Earth Engine (Python & JavaScript APIs), AutoCAD<br>*Hydrologic modeling*: HEC-RAS, SWMM<br>*Development environments*: VS Code, PyCharm, Jupyter Notebook, Google Colab, RStudio<br>*Surveying instruments*: Total Station, Robotic Theodolite, GNSS/GPS, UAVs, Leveling Instruments |
| **PERSONAL SKILLS** | Research synthesis & meta-analysis, scientific writing & peer publication, cross-disciplinary collaboration, teaching & academic mentoring. |

**AWARDS & HONORS**

| | | |
|---|---|---|
| 2026 | Student Travel Award ($700), College of Arts and Sciences<br>(*Selected via a competitive application process to present original research at the AAG Spring 25 Annual Meeting in San Francisco, CA.*) | USF, FL, USA |
| 2026 | Fred L. and Helene M. Tharp Endowed Scholarship ($5,400), USF – Summer 2026 | USF, FL, USA |
| 2026 | Outstanding Oral Presenter, Master's Category, Florida Society of Geographers | Gainesville, FL, USA |
| 2025 | Outstanding Poster Presenter, Master's Category, Florida Society of Geographers | Gainesville, FL, USA |
| 2023 | Certificate of National Service (Teaching and Research Assistant) | Kumasi, Ghana |
| 2022 | Excellent Student Award (College of Engineering, KNUST) | Kumasi, Ghana |
| 2021 | Recognition of Students' Academic Board Membership | Kumasi, Ghana |
| 2021 | Excellent Student Award (College of Engineering, KNUST) | Kumasi, Ghana |
| 2020 | Excellent Student Award (College of Engineering, KNUST) | Kumasi, Ghana |

**CERTIFICATIONS**

| | | |
|---|---|---|
| 2026 | Geographic Information Systems Professional (**GISP**)<br>**GIS Certification Institute (GISCI)**<br>**Certificate No: 162638**<br>**Earned: Jan 2026 – Expires: Jan 2029** | USA |
| 2025 | CITI Program: Social / Behavioral Investigators and Key Personnel | Tampa, FL, USA |
| 2024 | Professional Geospatialist (PG) Certification<br>Ghana Geospatial Society (GGS) | Kumasi, Ghana |
| 2024 | ESRI MOOC: GIS for Climate Action | |
| 2023 | ESRI MOOC: Spatial Data Science: The New Frontier in Analytics | |
| 2023 | ESRI MOOC: Transform AEC projects with GIS and BIM | |
| 2023 | ESRI MOOC: Cartography | |
| 2023 | ESRI MOOC: Going Places with Geospatial Analysis | |
| 2022 | Coursera: Machine Learning Pipelines with Azure ML Studio | |
| 2022 | Coursera: Azure Synapse SQL Pool – Implement Polybase | |
| 2022 | Geotech GIS: QGIS and Remote Sensing Online Training | Virtual (India) |
| 2020 | Certificate of Participation (GIS Day Celebration at KNUST) | Kumasi, Ghana |

**PUBLICATIONS**

August 2026 — **Conference submission:**
**Atta Amponsah, C***., Khan, T. I., Mumu, C. M. M., & Downs, J. (2026). Flood-risk-weighted emergency shelter optimization: Benchmarking Q-learning against exact p-median solvers. *2nd Workshop on Advances in Representation Learning for Earth Observation (REO-2), NeurIPS 2026*. Submitted.

June 2026 — **Thesis Publication:**
**Atta Amponsah, C***. (2026). The application of GIS-integrated ML models for flood prediction: A global meta-analysis and a comparative study of ML, AutoML, and HEC-RAS flood models (Publication No. 32738350) [Master's thesis, University of South Florida]. *ProQuest Dissertations & Theses Global*. https://www.proquest.com/docview/3358049991.

May 2026 — **Journal publication:**
**Atta Amponsah C***., Obosu P., Boateng C., Ofobi A., Waliba T. (2026). Performance Comparison of Machine Learning Algorithms in Land Cover Classification. *Geo-spatial Information Science*. https://doi.org/10.1080/10095020.2026.2667719. Taylor and Francis.

July 2026 — **Journal publication:**
**Atta Amponsah C***., Dixon B.[*], Qiang Y., Yan E. (2026). The Application of Geospatially-Integrated Machine Learning Models for Flood Prediction: A Review. *Journal of Hydrology*. https://doi.org/10.1016/j.jhydrol.2026.136016. Elsevier.

July 2026 — **Under Review:**
**Atta Amponsah C**.*, Khan T. I., Mumu C. M. M., Downs J. (2026). Flood-Risk-Weighted Emergency Shelter Optimization Using Reinforcement Learning and Exact p-Median Solvers. *Computers, Environments, and Urban Systems.* Elsevier.

May 2026 — **Under Review:**
**Atta Amponsah, C*.,** Dixon, B., Qiang, Y., Covarubias, S., Yan, E. (N/A). Cross-Event Generalizability of Machine Learning and AutoML Models for Hurricane-Induced Flood Susceptibility: Evaluation Against HEC-RAS Simulations. *Natural Hazards.* Springer Nature.

Nov 2024 **Conference Proceedings:**

**Atta Amponsah C***, Asiedu A. A., Achampong K. Y., Ayer J., Asante C. Y., Ziggah Y. Y. (2024). Mitigating Latent Land Conflicts in Ghana: The Role of Uniform Coordinate Systems. *Ghana Geospatial Society (GGS) 2nd Annual Geospatial Meeting & Conference Proceedings.*

## PROFESSIONAL MEETINGS AND PRESENTATIONS

June 2 - 5, 2026 **Conference attendance & presentation:** FL, USA

- **SWAN 2026 Annual Conference (Smart Water Networks Forum).** Attended as a delegate and volunteered with the Rising Smart Water Professionals (RiSWP) network. Tampa, FL. June 2-5.

  Co-presented a summary of key conference takeaways during the closing session. Tampa, FL. June 2-5.

  Co-authored a blog summarizing the conference highlights. https://swan-forum.com/publications/the-tools-are-ready-are-we/

March 2026 **Conference paper presentation:** CA, USA

- **Atta Amponsah C***., and Dixon B. (2026). HEC-RAS and Diverse AutoML-Optimized Suite of Machine Learning Approaches for Hurricane-Induced Flood Prediction: A Comparative Study. American Association of Geographers (AAG) 2026 Annual Meeting. San Francisco, CA. March 17-21. (*Recipient of the ASLO Student Travel Award ($750*).

March 2026 **Conference paper presentation:** CA, USA

- **Colarusso A**.*, Atta Amponsah C., and Dixon B. (2026). Urban Flood and Damage Prediction with Real-time Rainfall Data and Flood Depth derived from a Camera Network Using a Convolutional Neural Network. American Association of Geographers (AAG) 2026 Annual Meeting. San Francisco, CA. March 17-21.

Feb 2026 **Conference paper presentation:** FL, USA

- **Atta Amponsah C***. and Dixon B. (2026). Comparison of CNN, RF, and XGBoost for Flood Prediction: Insights from Hurricanes Helene and Milton. Florida Society of Geographers 61st Annual Meeting & Conference, 2026. Gainesville, FL. February 20-22

June 2025 **Conference attendance:** FL, USA

- Florida Floodplain Managers Association (FFMA) 2025 Annual Conference. Cape Coral, FL. June 24-27.

May 2025 **Conference poster presentation:** FL, USA

- **Atta Amponsah C***. and Dixon B. (2025). The Application of GIS-Integrated Machine Learning in Flood Prediction: A Meta-analysis and A Comparative Modeling of CNN And HEC-RAS Flood Modeling. The American Society of Photogrammetry and Remote Sensing (ASPRS) Spring Geospatial Workshop 2025. Gainesville, FL. May 8.

Feb 2025 **Conference poster presentation:** FL, USA

- **Atta Amponsah C***. and Dixon B. (2025). Machine Learning Applications Integrated with GIS for Flood Prediction. Florida Society of Geographers 61st Annual Meeting & Conference, 2025. Gainesville, FL. February 20-22.

## REFERENCES

Available upon request